

Mathematical Statistics Notes

Structured from Handwritten Notes

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5 Multivariate Distribution

5.1 Joint Distributions

- **Discrete:** The joint probability is defined as $f_{X,Y}(x, y) = P(X = x, Y = y)$ [cite: 239, 240]. The marginal PDF is $f_X(x) = \sum_y f_{X,Y}(x, y)$ [cite: 243]. The CDF is $F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$ [cite: 245].
- **Continuous:** For continuous random variables, $\iint_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$ [cite: 293]. The marginal PDF is computed via integration: $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$ [cite: 307].

5.2 Expectation, Covariance, and Independence

- **Expected Values:** $E[g(X, Y)] = \iint g(x, y) f_{X,Y}(x, y) dx dy$ [cite: 309]. Furthermore, $E[aX + bY + c] = aE[X] + bE[Y] + c$ [cite: 311].
- **Independence:** Random variables X and Y are independent if $f_{X,Y}(x, y) = f_X(x)f_Y(y)$ for all x, y [cite: 318]. If independent, $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$ and $Var(X + Y) = Var(X) + Var(Y)$ [cite: 327, 328].
- **Covariance:** $Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - E[X]E[Y]$ [cite: 330, 331].
- **Correlation Coefficient:** $\rho = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$ [cite: 338].

Conditional Distributions

The conditional PDF of Y given $X = x$ is defined as:

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)} \quad [\text{cite: 359}]$$

The conditional variance is defined as $Var[Y|X = x] = E[Y^2|X = x] - (E[Y|X = x])^2$ [cite: 380].

6 Functions of Multivariate R.V.

6.1 Transformation Methods

- **CDF Method:** For $W = h(X, Y)$, the CDF is $F_W(w) = P(h(X, Y) \leq w) = \iint_{h(x, y) \leq w} f_{X,Y}(x, y) dx dy$ [cite: 412, 414].
- **Transformation Method (Jacobian):** For a transformation $Y = u(X)$, the joint PDF is $f_Y(y) = f_X(x)|J|$ where J is the Jacobian determinant of the transformation matrix[cite: 426, 427, 428].
- **MGF Method:** Uses Moment Generating Functions, e.g., $M_{U,W}(t_1, t_2) = E[e^{t_1 U + t_2 W}]$ [cite: 444, 445].

6.2 Distributions of Sums & Sample Statistics

- **Sums:** $E[\sum_{i=1}^k X_i] = \sum_{i=1}^k E[X_i]$. If independent, $Var[\sum_{i=1}^k X_i] = \sum_{i=1}^k Var[X_i]$ [cite: 449, 458].
- **Convolution Formula:** If X, Y are independent, the PDF of $S = X + Y$ is $f_S(s) = \int_{-\infty}^{\infty} f_X(x)f_Y(s-x)dx$ [cite: 466, 468].

Sample Distributions and Order Statistics

Sample Statistics:

- Sample Mean: $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, with $E[\bar{X}] = \mu$ and $Var[\bar{X}] = \sigma^2/n$ [cite: 495, 497].
- Sample Variance: $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ [cite: 498].

Order Statistics: For a sample size n , the 1st order statistic is $Y_1 = \min(X_1 \dots X_n)$ and the n -th is $Y_n = \max(X_1 \dots X_n)$ [cite: 524]. The PDF of the k -th order statistic is:

$$f_{Y_k}(y) = \frac{n!}{(k-1)!(n-k)!} [F(y)]^{k-1} [1-F(y)]^{n-k} f(y) \quad [\text{cite: 537}]$$

7 Estimation

7.1 Point Estimation

- A point estimator uses sample statistics to estimate a population parameter[cite: 546].
- Estimator for σ^2 is S^2 , which is unbiased: $E[S^2] = \sigma^2$ [cite: 548, 549].
- Estimator for proportion p is $\hat{p} = \frac{X}{n}$ with $E[\hat{p}] = p$ and $Var(\hat{p}) = \frac{p(1-p)}{n}$ [cite: 552, 553, 554].

7.2 Confidence Intervals (CI)

100(1- α)% Confidence Intervals

For a single population:

- μ (σ known): $\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ [cite: 557].
- μ (σ unknown): $\bar{X} \pm t_{n-1, \alpha/2} \frac{S}{\sqrt{n}}$ [cite: 567].
- Proportion p : $\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ [cite: 571].
- Variance σ^2 : $[\frac{(n-1)S^2}{\chi_{\alpha/2}^2(n-1)}, \frac{(n-1)S^2}{\chi_{1-\alpha/2}^2(n-1)}]$ [cite: 566].

8 Hypothesis Testing

8.1 Fundamental Concepts

- **Steps:** 1) Formulate null (H_0) & alternative (H_a) hypotheses; 2) Determine test statistics; 3) Describe assumptions; 4) Determine critical region[cite: 580, 581, 582].
- **Errors & Power:**
 - Type I Error: $P(\text{reject } H_0 | H_0 \text{ is true}) = \alpha$ [cite: 605].
 - Type II Error: $P(\text{not reject } H_0 | H_0 \text{ is false}) = \beta$ [cite: 606].
 - Power of Test: $P(\text{reject } H_0 | H_0 \text{ is false}) = 1 - \beta$ [cite: 607].
- **p-value:** The probability of obtaining a result equal to or "more extreme" than what was actually observed, assuming H_0 is true[cite: 615]. If $p\text{-value} \leq \alpha$, H_0 can be rejected[cite: 616].

Tests for One and Two Populations

Single Population Tests:

- For μ (σ known): $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$ [cite: 619, 631].
- For μ (σ unknown): $T_{n-1} = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t(n-1)$ [cite: 620, 632].
- For p : $Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \sim N(0, 1)$ [cite: 644].

Two Population Tests:

- Difference in means ($\mu_1 - \mu_2$, σ_1, σ_2 known): $Z = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ [cite: 658].
- Ratio of variances (σ_1^2/σ_2^2): $F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1)$ [cite: 671].